

# Supplementary Material for “Reproducible model-based AI planning for county-scale farmland consolidation in fragmented mountain landscapes”

Ning Zhou and Xiang Jing

## S1 In-house model-free DRL baseline configuration

The Centralised PPO and Multi-Agent Reinforcement Learning (MARL) baselines reported in main-text Section 4 (and used as the fairness anchor for the contrastive-MPC results) were trained in-house under the following protocol.

**Environment.** Both baselines act on the same Bishan County county-level environment (`CountyLevelEnv`) used by the contrastive MPC pipeline: 13 townships, 2,600 spatial blocks aggregating 52,515 cadastral parcels. The observation is the concatenation of 2,600 blocks  $\times$  17 block-level features plus 12 global features. The action space is a categorical choice over the 2,600 blocks, with the same MaskablePPO-style action mask described in main-text Methods.

**Reward.** Identical to the main-text reward (Equation 1):  $w_S = 4,000$ ,  $w_C = 500$ ,  $w_A = 1,500$ ,  $b_B = 5$ ,  $p_A = 2,000$ . No method-specific re-tuning was performed, so cross-method comparisons reflect the learning procedure rather than reward engineering.

**Centralised PPO.** A single MaskablePPO agent with our `ParcelScoringPolicy` architecture (block-wise scoring MLP  $[64, 32] \rightarrow 1$ , global-feature value network). Training: 5 random seeds,  $1.5 \times 10^6$  environment steps each,  $\gamma = 0.995$ ,  $\lambda = 0.95$ , learning rate  $1 \times 10^{-3}$  linear-decayed, entropy coefficient 0.01, clip range 0.2, batch size 4,096, mini-batch size 256, 10 epochs per update. Wall-time: 8–12 GPU-hours per seed on a single A100. Cross-seed slope improvement:  $-0.79\% \pm 0.36\%$ , with 2/5 seeds collapsing into degenerate trajectories.

**Multi-Agent Reinforcement Learning (MARL).** A 13-agent variant in which each township is a separate MaskablePPO agent with shared parameters and a centralised value function over the global state, trained jointly. Same hyperparameters as Centralised PPO. 5 seeds, all stable. Cross-seed slope improvement:  $-0.81\% \pm 0.10\%$  ( $CV \approx 12\%$ ).

**Why these are the relevant baselines.** Both baselines are task-faithful: same environment, same action space, same reward function, same evaluation protocol (5-episode deterministic rollout) as the contrastive MPC pipeline. The wall-time and seed-failure characteristics motivate the search for a less compute-intensive learned-model alternative; the slope-improvement magnitudes ( $-0.8\%$  range) provide a non-trivial reference against which the contrastive-MPC result ( $-1.289\% \pm 0.079\%$  on Bishan) is then compared.

## S2 Real-county initial state

The two real-county datasets anchoring the cadastral evaluation in main-text Section 2 are summarised in Table S1. All values are extracted from the Third National Land Survey records and aggregated to spatial blocks (the planning units used by the model); initial-state metrics are computed from the cadastral data before optimisation.

Table S1: Real-county initial state for Bishan District (Chongqing) and Neijiang Dongxing District (Sichuan).

|                                     | Bishan, Chongqing | Neijiang Dongxing, Sichuan |
|-------------------------------------|-------------------|----------------------------|
| Townships                           | 13                | 29                         |
| Cadastral parcels                   | 52,515            | 76,376                     |
| Spatial blocks (planning units)     | 2,600             | 3,711                      |
| Initial farmland slope (deg)        | 9.62              | 10.55                      |
| Initial contiguity index            | 3.59              | 2.63                       |
| Initial <i>baimu fang</i> count     | 109               | 384                        |
| Initial <i>baimu fang</i> area (ha) | 46,844            | 74,342                     |

## S3 Bishan replication: cross-pipeline reconciliation

The main text reports Bishan slope-improvement numbers from two cadastral pipelines, plus a single GIS-only on-disk verification anchor and an open-source CLI reproduction. We tabulate them here for transparent cross-reference. All four entries evaluate the same underlying contrastive-MPC pipeline ( $\lambda_{\text{rank}}=5.0$ ,  $H=5$ ,  $K=50$ , greedy continuation,  $\gamma=0.99$ ); none is a re-tuning of the algorithm. The shipped ensemble used in the package-side rows is the public release `paper/checkpoints/bishan/shipped_onnx/` (3-member ensemble,  $\lambda_{\text{rank}}=5.0$ ); the lab-pipeline ensembles are the public release `paper/checkpoints/bishan/contrastive_5seed/` (five 3-member ensembles, seeds 0–4).

**Why two pipelines coexist.** The lab pipeline was used to train the in-house Centralised PPO and multi-agent baselines and is preserved here only as the matched-cadastre comparison reported in main-text §4. The package-side prepare is the public-facing toolchain shipped with the repository and the cadastre on which the headline result is computed; it differs from the lab pipeline in two respects relevant to the present numbers: (a) it computes block aggregation from the DLTB shapefile directly rather than from a precomputed `DLTB_with_slope.gpkg`, so the block boundaries differ slightly (2,640 vs. 2,600 blocks); (b) it preserves the true non-square pixels of geographic-CRS DEMs (commit `dc3a01c`), which fixes a slope-underestimation bug in the lab pipeline and exposes a  $\sim 1.2^\circ$  taller mean initial slope. The two pipelines therefore present the same algorithm operating on slightly different prepared inputs; the gap is morphological rather than algorithmic. The 82-parcel difference between the env-side count (53,004) and the verification-side count (53,086) is auxiliary: the verification script keeps every parcel with a slope value and a farm/forest `ORIG_DLBM` regardless of township-registry membership, while the env enforces township membership; the additional 82 parcels all carry `CHG_FLAG = 0` and contribute zero to every delta.

**Cross-pipeline reproduction by the open-source toolchain.** An end-to-end reproduction of all five §5 metrics on the package-side prepare from raw DLTB shapefiles plus DEM

Table S2: Bishan replication across cadastral pipelines. The on-disk GIS-only verification of the package-side prepare ( $-2.001\%$ ) is the headline anchor of main-text Section 4. The five-ensemble lab-pipeline run ( $-1.289 \pm 0.079\%$ ) is the matched-cadaastre comparison against the in-house Centralised PPO and multi-agent baselines, both of which were trained on the lab pipeline before the prepare-step correction. The CLI five-episode and the open-source end-to-end reproduction are reported as deployment-side checks. The pipeline-level gap ( $\sim 0.7$  pp on slope improvement) is morphological: the corrected geographic-CRS slope step in the package-side prepare exposes a  $\sim 1.2^\circ$  taller initial mean slope ( $9.62^\circ$  vs  $8.41^\circ$ , a 14% underestimate on the lab pipeline), leaving more headroom for reduction within the same 100-step episode budget.

| Row                            | GIS-only verification  | Lab pipeline ( $n=5$ ens.) | CLI 5-ep             | Open-source repro    |
|--------------------------------|------------------------|----------------------------|----------------------|----------------------|
| Slope change %                 | $-2.001$               | $-1.289 \pm 0.079$         | $-1.544 \pm 0.041$   | $-1.7384 \pm 0.024$  |
| $\Delta$ Contiguity            | $+0.01275$             | $+0.0160 \pm 0.0016$       | not reported         | $+0.01442 \pm 0.001$ |
| $\Delta$ Baimu count           | $+8$                   | $+3.4 \pm 1.0$             | not reported         | $+3.40 \pm 1.36$     |
| $\Delta$ Baimu (ha)            | $-473.30$              | $-312.2 \pm 34.3$          | not reported         | $-478.64 \pm 35.0$   |
| $\Delta$ Total cultivated (ha) | $-505.23$              | not reported               | not reported         | not reported         |
| Replication unit               | 1 ens. $\times$ 1 ep   | 5 ens. $\times$ 1 ep       | 1 ens. $\times$ 5 ep | 5 ens. $\times$ 5 ep |
| Variance estimator             | 2-of-3 subset ablation | cross-ensemble             | cross-episode        | cross-ensemble       |
| Prepare pipeline               | package-side           | lab (legacy CRS)           | package-side         | package-side         |
| Initial slope $\bar{\sigma}_0$ | $9.62^\circ$           | $8.41^\circ$               | $9.62^\circ$         | $9.62^\circ$         |
| Blocks $N_{\text{blocks}}$     | 2,640                  | 2,600                      | 2,640                | 2,640                |
| Parcel set                     | 53,086 (verification)  | 52,515                     | 53,004 (env)         | 53,004 (env)         |
| Anchor in main text            | §4 + Methods §10.4     | §4 (DRL comparison)        | §6 (toolchain)       | SI §S2 only          |
| Source artefact                | GIS-only script        | contrastive_5seed/         | shipped_onnx/        | repro_artifacts/     |

tiles (`paper/repro_artifacts/macos_2026-05-29/bishan_5seed.json`) achieves slope  $-1.7384 \pm 0.024\%$ , contiguity  $+0.01442 \pm 0.001$ , baimu count  $+3.40 \pm 1.36$ , baimu area  $-478.64 \pm 35.0$  ha. This bracket reproduces the qualitative cross-county finding (Bishan loses qualifying area; Dongxing gains it) end-to-end without access to the lab pipeline. A separate package-CLI 5-episode greedy run on the shipped ensemble (`runs/bishan/mpc_output/mpc_summary.json`) yields slope  $-2.0392 \pm 0.0010\%$ , baimu area  $-499.12 \pm 0.17$  ha, with the  $\sim 0.04$  pp offset from the on-disk verification anchor of  $-2.001\%$  attributable to ensemble-determinism path differences across re-runs of the deployment dataset. The full reproduction recipe is documented in `docs/REPRODUCE.md`; the per-seed JSON outputs are released alongside the toolchain.

## S4 Statistical tests reported in the main text

Table S3 consolidates the inline statistical tests cited in the main text. Tests against in-house DRL baselines (Centralised PPO, MARL) use Welch’s two-sample  $t$ -test on summary statistics (the per-seed run logs of the in-house baselines are not redistributed; means and sample standard deviations are reported in §S1). Tests between contrastive ( $\lambda=5$ ) and MSE-only ( $\lambda=0$ ) ensembles on Buchanan are paired tests over  $n=5$  matched ensemble training seeds, using both Welch’s paired  $t$  and a Wilcoxon signed-rank test as a non-parametric robustness check, plus a TOST equivalence test at  $\Delta = 0.5$  reward units ( $\sim 0.22\%$  of mean reward). All tests are two-sided unless stated otherwise. Effect sizes are Cohen’s  $d$  (independent) or  $d_z$  (paired); pooled standard deviations follow Hedges & Olkin’s convention.

The Buchanan paired  $t$ -test result ( $p=0.060$ , marginally outside the conventional 0.05 supe-

Table S3: All inline statistical tests reported in main text §5 and §6, with raw inputs and effect sizes. Test types: W2 = Welch’s two-sample  $t$ ; PT = paired  $t$ ; WSR = Wilcoxon signed-rank; TOST = two-one-sided-tests equivalence at the stated margin  $\Delta$ . Cohen’s effect-size thresholds (Cohen 1988):  $|d|>0.8$  large,  $|d|>1.2$  very large.

| Comparison                                                               | Group<br>(mean $\pm$ sd,<br>$n$ ) | 1      | Group<br>(mean $\pm$ sd,<br>$n$ ) | 2      | Test | Statistic                                                        | $p$         | Effect size                |
|--------------------------------------------------------------------------|-----------------------------------|--------|-----------------------------------|--------|------|------------------------------------------------------------------|-------------|----------------------------|
| Bishan slope: contrastive vs Centralised PPO                             | $-1.289\pm0.079\%$<br>(5)         |        | $-0.79\pm0.36\%$<br>(5)           |        | W2   | $t = -3.01$ , $df=4.5$                                           | 0.034       | $d = -1.90$ (very large)   |
| Bishan slope: contrastive vs MARL                                        | $-1.289\pm0.079\%$<br>(5)         |        | $-0.81\pm0.10\%$<br>(5)           |        | W2   | $t = -8.01$ , $df=7.9$                                           | $< 10^{-4}$ | $d = -5.07$ (very large)   |
| Buchanan reward: contrastive ( $\lambda=5$ ) vs MSE-only ( $\lambda=0$ ) | $229.39\pm0.030$<br>(5)           |        | $229.20\pm0.157$<br>(5)           |        | PT   | $t = +2.61$ , $df=4$                                             | 0.060       | $d_z = +1.17$ (very large) |
| Buchanan equivalence at $\Delta = \pm 0.5$ reward units                  | —                                 |        | —                                 |        | TOST | $t_{\text{lower}} = +9.25$ , $t_{\text{upper}} = -4.04$ , $df=4$ | 0.008       | equivalence asserted       |
| Buchanan reward (non-parametric robustness)                              | median<br>(5)                     | 229.40 | median<br>(5)                     | 229.26 | WSR  | $W=0.0$                                                          | 0.063       | —                          |

riority threshold) and the TOST equivalence result ( $p_{\text{TOST}}=0.008$  at a  $\pm 0.22\%$ -of-mean-reward margin) jointly support the main-text reading: a small, reproducible, but operationally negligible reward advantage of the contrastive variant in the regime where MSE training already acquires the ranking signal as a by-product. Whether the small superiority signal would survive at  $n > 5$  is unresolved at the present sample size; the equivalence claim is the conservative reading the main text adopts. Source per-seed reward data: `runs/restoration/buchanan_va/5seed_results.json` (contrastive,  $\lambda=5$ ) and `5seed_lam0_results.json` (MSE-only,  $\lambda=0$ ); the `paper/checkpoints/restoration/buchanan_va/` directory holds the corresponding ensemble weights for both conditions.

## S5 MPC optimisation ablation on the MSE-only ensemble (Supplementary Table S1)

Table S4 below reports the full MPC optimisation ablation cited in main-text Section 3 (paragraph beginning “Four independent operational failures”). All variants use the same MSE-only world-model ensemble (Bishan District, 5-episode evaluation) and modify a single dimension of the planner. None exceeds the random-sampling baseline; the catastrophic failure of greedy continuation under the MSE-only ensemble (slope  $+0.029\%$ , wrong direction) is the variant that contrastive training subsequently rescues into the strongest variant tested (main-text Table 1, the  $\lambda=5.0$  rows).

## S6 Synthetic-benchmark secondary metrics (Tables S2–S4)

Table S5 reports the slope-improvement profile across the seven synthetic landscapes, summarised in main-text Section 5. The three secondary metrics referenced in main-text Section 5 ( $\Delta_{\text{contiguity}}$ ,  $\Delta_{\text{baimu fang count}}$ , and  $\Delta_{\text{baimu fang area}}$ , all under the same 50-pair swap budget and  $n=5$  seeds per cell) are tabulated below the slope profile. Together these summarise the full  $5 \times 7 \times 5 = 175$ -cell sweep. Aggregation script: `scripts/si_tables_v7.py` in the released code repository.

Table S4: MPC optimisation ablation across four dimensions, on the MSE-only world-model ensemble (Bishan District). All variants start from the optimal  $H=5$ ,  $K=50$  random-reward configuration and modify one dimension. None exceeds the random-sampling baseline; aggressive exploitation (greedy continuation) actively reverses the slope direction. The diagnostic shows that the failure is structural to the learned model rather than to the search procedure.

| Dimension      | Configuration            | Slope %            | Reward | Diagnosis           |
|----------------|--------------------------|--------------------|--------|---------------------|
| Continuation   | Random (baseline)        | $-0.209 \pm 0.008$ | -18.52 | —                   |
|                | Greedy                   | $+0.029 \pm 0.005$ | -95.29 | catastrophic        |
| Scoring        | Reward (baseline)        | $-0.209 \pm 0.008$ | -18.52 | —                   |
|                | Slope-only               | -0.001             | -98.95 | zero discrimination |
| Terminal value | No value (baseline)      | $-0.209 \pm 0.008$ | -18.52 | —                   |
|                | Value-guided ( $v=1.0$ ) | $-0.172 \pm 0.042$ | -27.96 | no improvement      |
| Search budget  | $K=50$ (baseline)        | $-0.209 \pm 0.008$ | -18.52 | —                   |
|                | $K=100$                  | $-0.167 \pm 0.053$ | -8.94  | no improvement      |
|                | $K=50$ , $n_r=3$         | $-0.171 \pm 0.046$ | -28.33 | no improvement      |

Table S5: Slope-improvement profile of five methods across the seven synthetic landscapes,  $n=5$  seeds per cell (175 cells total). Values are mean  $\pm$  sample standard deviation of the relative slope change in percent (negative is better). The rightmost column reports median wall-time per cell (seconds for fast methods, hours for learning-based). Best mean per row in bold. Contrastive MPC is the strongest method on the five planning-favourable presets (the two real-data clones and the three medium-large fragmented presets); the genetic algorithm wins on `plain_small_cons`, where the small action space lets a population search cover the optimum within budget. Numbers regenerated from `benchmark/results/` via `scripts/aggregate_results.py`.

| Preset ( $n_{\text{blocks}}$ ) | Random             | Greedy             | GA                                   | PPO                | MPC                                  | Wall (MPC) |
|--------------------------------|--------------------|--------------------|--------------------------------------|--------------------|--------------------------------------|------------|
| bishan_clone (2,600)           | $-0.026 \pm 0.003$ | $-0.052 \pm 0.009$ | $-0.060 \pm 0.009$                   | $-0.074 \pm 0.015$ | <b><math>-0.089 \pm 0.014</math></b> | 4.1h       |
| neijiang_clone (3,700)         | $-0.011 \pm 0.002$ | $-0.025 \pm 0.004$ | $-0.026 \pm 0.005$                   | $-0.036 \pm 0.011$ | <b><math>-0.040 \pm 0.007</math></b> | 6.1h       |
| plain_small_cons (800)         | $-0.544 \pm 0.072$ | $-0.996 \pm 0.268$ | <b><math>-2.757 \pm 0.203</math></b> | $-2.234 \pm 0.264$ | $-2.226 \pm 0.220$                   | 2.0h       |
| plain_large_cons (4,500)       | $-0.071 \pm 0.015$ | $-0.128 \pm 0.020$ | $-0.309 \pm 0.018$                   | $-0.419 \pm 0.060$ | <b><math>-0.424 \pm 0.032</math></b> | 10.9h      |
| plain_medium_frag (2,600)      | $-0.132 \pm 0.028$ | $-0.310 \pm 0.080$ | $-0.651 \pm 0.029$                   | $-0.773 \pm 0.076$ | <b><math>-0.805 \pm 0.069</math></b> | 6.3h       |
| mixed_medium_frag (2,600)      | $-0.078 \pm 0.018$ | $-0.173 \pm 0.026$ | $-0.405 \pm 0.025$                   | $-0.414 \pm 0.076$ | <b><math>-0.431 \pm 0.033</math></b> | 5.3h       |
| hilly_small_cons (800)         | $-0.090 \pm 0.028$ | $-0.152 \pm 0.012$ | $-0.295 \pm 0.050$                   | $-0.260 \pm 0.035$ | <b><math>-0.271 \pm 0.047</math></b> | 1.1h       |

Table S6: Synthetic-benchmark contiguity change ( $\Delta\text{Contiguity}$ , absolute units). Mean  $\pm$  standard deviation across  $n=5$  seeds. Higher (more positive) is better; per-row best in bold.

| Preset ( $n_{\text{blocks}}$ ) | Random              | Greedy              | GA                                    | PPO                                   | MPC                                   |
|--------------------------------|---------------------|---------------------|---------------------------------------|---------------------------------------|---------------------------------------|
| bishan_clone (2,600)           | $0.0016 \pm 0.0007$ | $0.0009 \pm 0.0006$ | <b><math>0.0253 \pm 0.0023</math></b> | $0.0027 \pm 0.0005$                   | $0.0032 \pm 0.0009$                   |
| neijiang_clone (3,700)         | $0.0014 \pm 0.0004$ | $0.0012 \pm 0.0001$ | <b><math>0.0159 \pm 0.0005</math></b> | $0.0015 \pm 0.0004$                   | $0.0023 \pm 0.0005$                   |
| plain_small_cons (800)         | $0.0057 \pm 0.0023$ | $0.0036 \pm 0.0010$ | $0.0018 \pm 0.0057$                   | <b><math>0.0069 \pm 0.0014</math></b> | $0.0068 \pm 0.0015$                   |
| plain_large_cons (4,500)       | $0.0010 \pm 0.0001$ | $0.0008 \pm 0.0002$ | $0.0010 \pm 0.0003$                   | <b><math>0.0014 \pm 0.0002</math></b> | $0.0012 \pm 0.0003$                   |
| plain_medium_frag (2,600)      | $0.0018 \pm 0.0004$ | $0.0018 \pm 0.0005$ | $0.0005 \pm 0.0006$                   | $0.0022 \pm 0.0003$                   | <b><math>0.0026 \pm 0.0004</math></b> |
| mixed_medium_frag (2,600)      | $0.0022 \pm 0.0006$ | $0.0018 \pm 0.0003$ | <b><math>0.0059 \pm 0.0016</math></b> | $0.0031 \pm 0.0007$                   | $0.0032 \pm 0.0008$                   |
| hilly_small_cons (800)         | $0.0058 \pm 0.0020$ | $0.0067 \pm 0.0017$ | <b><math>0.0734 \pm 0.0055</math></b> | $0.0092 \pm 0.0020$                   | $0.0095 \pm 0.0013$                   |

Table S7: Synthetic-benchmark large-patch count change ( $\Delta baimu\ fang$  count,  $\geq 6.67$  ha qualifying patches gained per episode). Mean  $\pm$  standard deviation across  $n=5$  seeds. Higher is better; per-row best in bold.

| Preset ( $n_{\text{blocks}}$ ) | Random         | Greedy         | GA                            | PPO                           | MPC                           |
|--------------------------------|----------------|----------------|-------------------------------|-------------------------------|-------------------------------|
| bishan_clone (2,600)           | 0.0 $\pm$ 2.2  | -0.2 $\pm$ 2.6 | -3.8 $\pm$ 4.4                | <b>9.4<math>\pm</math>2.6</b> | 9.2 $\pm$ 2.3                 |
| neijiang_clone (3,700)         | 0.6 $\pm$ 0.9  | -1.4 $\pm$ 1.5 | -4.4 $\pm$ 3.2                | <b>6.4<math>\pm</math>2.1</b> | 4.6 $\pm$ 3.4                 |
| plain_small_cons (800)         | 0.0 $\pm$ 0.0  | 0.2 $\pm$ 1.1  | <b>0.8<math>\pm</math>2.0</b> | -0.4 $\pm$ 2.5                | 0.4 $\pm$ 2.1                 |
| plain_large_cons (4,500)       | 0.0 $\pm$ 0.0  | 0.0 $\pm$ 0.0  | 0.2 $\pm$ 0.4                 | <b>0.8<math>\pm</math>0.8</b> | 0.0 $\pm$ 0.7                 |
| plain_medium_frag (2,600)      | 0.6 $\pm$ 0.9  | -0.6 $\pm$ 1.3 | 1.4 $\pm$ 1.8                 | <b>1.6<math>\pm</math>1.1</b> | 0.4 $\pm$ 1.3                 |
| mixed_medium_frag (2,600)      | -0.2 $\pm$ 1.3 | 0.0 $\pm$ 1.2  | <b>1.2<math>\pm</math>2.2</b> | -1.0 $\pm$ 2.2                | -2.4 $\pm$ 1.1                |
| hilly_small_cons (800)         | -1.0 $\pm$ 0.7 | -0.4 $\pm$ 0.9 | -3.2 $\pm$ 2.2                | 3.4 $\pm$ 1.7                 | <b>9.8<math>\pm</math>1.5</b> |

Table S8: Synthetic-benchmark large-patch area change ( $\Delta baimu\ fang$  area, ha). Mean  $\pm$  standard deviation across  $n=5$  seeds. The sign of this metric is morphology-dependent (see main-text Section 5 discussion of the Bishan vs. Neijiang reversal); per-row best (most positive) in bold for reference.

| Preset ( $n_{\text{blocks}}$ ) | Random          | Greedy          | GA                               | PPO                              | MPC                              |
|--------------------------------|-----------------|-----------------|----------------------------------|----------------------------------|----------------------------------|
| bishan_clone (2,600)           | 9.8 $\pm$ 11.1  | 11.9 $\pm$ 15.0 | 155.3 $\pm$ 40.3                 | 134.7 $\pm$ 26.1                 | <b>177.5<math>\pm</math>31.7</b> |
| neijiang_clone (3,700)         | 15.2 $\pm$ 9.2  | 27.8 $\pm$ 16.4 | <b>190.8<math>\pm</math>33.2</b> | 122.4 $\pm$ 44.8                 | 167.6 $\pm$ 23.0                 |
| plain_small_cons (800)         | 20.2 $\pm$ 8.9  | 14.7 $\pm$ 16.2 | 11.5 $\pm$ 19.5                  | 19.0 $\pm$ 25.3                  | <b>30.5<math>\pm</math>17.5</b>  |
| plain_large_cons (4,500)       | 9.2 $\pm$ 8.9   | 39.7 $\pm$ 13.3 | 12.8 $\pm$ 13.4                  | 40.1 $\pm$ 55.3                  | <b>40.3<math>\pm</math>15.3</b>  |
| plain_medium_frag (2,600)      | 22.8 $\pm$ 10.5 | 38.5 $\pm$ 22.8 | 34.2 $\pm$ 34.3                  | <b>54.0<math>\pm</math>40.1</b>  | 52.8 $\pm$ 26.8                  |
| mixed_medium_frag (2,600)      | 14.2 $\pm$ 7.3  | 28.4 $\pm$ 13.6 | 60.7 $\pm$ 24.8                  | <b>106.8<math>\pm</math>24.2</b> | 102.1 $\pm$ 13.1                 |
| hilly_small_cons (800)         | 21.6 $\pm$ 14.2 | 4.6 $\pm$ 13.9  | <b>167.7<math>\pm</math>41.2</b> | 100.3 $\pm$ 30.3                 | 131.9 $\pm$ 43.5                 |

## S7 Per-seed Bishan multi-objective profile

Main-text Section 4 reports the matched-DRL-comparison mean of all four contrastive-MPC primary objectives on Bishan under the lab pipeline ( $\lambda_{\text{rank}}=5.0$ , greedy,  $H=5$ ,  $K=50$ , 5 independently trained ensembles  $\times$  1 episode each). The per-seed values that produce that mean are listed in Table S9. Initial state on the lab pipeline: slope  $8.41^\circ$  ( $9.62^\circ$  on the package-side prepare under the corrected geographic-CRS slope step, main-text Methods *Prepare-step slope correction*), contiguity 3.585, 109 *baimu fang* totalling 46,844 ha. Slope and contiguity changes are both directionally correct on every seed; the count of large patches ( $\geq 6.67$  ha) increases by +3.4 on average; total qualifying area decreases by  $\sim 312$  ha because the agent consolidates many small patches into a smaller number of larger, contiguous patches—the deliberate trade-off discussed in main-text Discussion. The corresponding package-side anchor (single canonical episode, GIS-only verification) loses  $-473$  ha of qualifying area and  $-505$  ha of total cultivated area; see §S3 for the cross-pipeline reconciliation and §S11 for the policy-band breakdown.

Table S9: Per-seed multi-objective profile of contrastive MPC on Bishan across 5 independently trained ensembles, 1 episode each, on the lab pipeline (initial slope  $8.41^\circ$ ). The DRL baselines used in the comparison were trained on the same pipeline; see §S3 for the cross-pipeline reconciliation against the package-side  $-2.001\%$  verification anchor.

| Seed        | Slope %                              | $\Delta$ Contiguity                    | $\Delta$ Baimu #                 | $\Delta$ Baimu (ha)                 |
|-------------|--------------------------------------|----------------------------------------|----------------------------------|-------------------------------------|
| 0           | -1.4235                              | +0.0150                                | +3                               | -324.9                              |
| 1           | -1.2668                              | +0.0165                                | +3                               | -283.6                              |
| 2           | -1.3250                              | +0.0159                                | +2                               | -372.7                              |
| 3           | -1.2264                              | +0.0139                                | +5                               | -278.2                              |
| 4           | -1.2018                              | +0.0186                                | +4                               | -301.9                              |
| <b>Mean</b> | <b><math>-1.289 \pm 0.079</math></b> | <b><math>+0.0160 \pm 0.0016</math></b> | <b><math>+3.4 \pm 1.0</math></b> | <b><math>-312.2 \pm 34.3</math></b> |

## S8 Toolchain pre-flight checks

Tool 5 of the ArcGIS Pro toolchain (main-text Section 8) validates the user environment before any planning step is allowed to run. The complete list of checks is:

1. **Python interpreter:** ArcGIS Pro 3.6+ `arcgispro-py3` or compatible Python 3.10+ environment.
2. **Spatial Analyst extension:** licence checked out (required by Tool 1 for slope derivation).
3. **Required packages:** `gymnasium` ( $\geq 0.29$ ), `libpysal` ( $\geq 4.7$ ), `onnxruntime` ( $\geq 1.16$ ), `numpy`, `pandas`, `scipy`.
4. **ONNX-ensemble shape compatibility:** if a pre-trained ensemble is provided, the baked-in  $N_{\text{blocks}}$  matches the prepared dataset.
5. **Coordinate reference system:** the input shapefile’s CRS resolves to a metric projection (default EPSG:32648; user-overridable).
6. **Disk space:** at least 5 GB free in the prepared-data directory.

Tool 5 prints an [OK] or [FAIL] verdict for each item and refuses to proceed with downstream tools if any required item fails. The 70-second smoke test on the bundled 2-township subset of Neijiang Dongxing District exercises every check end to end and is the recommended onboarding action for new users of the toolchain.

## S9 Data and code release

The synthetic benchmark, the pre-trained Bishan ensemble, the ArcGIS Pro toolbox source, and the aggregation/sweep scripts referenced throughout this paper are released under the licences indicated in the main-text Data and Code Availability statements. Repository structure at the public release:

- `LandUseOptimization_P9.pyt`: ArcGIS Pro Python Toolbox manifest.
- `farmland_mpc/`: pure-Python core (loss functions, ensemble training, MPC planner, ONNX export).
- `benchmark/`: synthetic generator, presets, baseline runners, sweep manifest, and the 175-cell results.
- `docs/BENCHMARK_RESULTS.md`: per-(preset, method) aggregated results derived from `benchmark/results/`.
- `scripts/aggregate_results.py`, `scripts/si_tables_v7.py`: aggregation and SI-table generators.
- `environment.yml`, Colab demo notebook, integration smoke test.

## S10 Verification stack: full per-episode data (Tables S5–S8)

The main-text Methods sections (Independent GIS-grounded verification, Direct measurement of ensemble 1-step prediction accuracy, and Replacing the ensemble with the true environment) summarise four of the five independent verification layers underwriting the Bishan headline result (the fifth, Neijiang cross-county replication, is in main-text Section 4). Tables S10–S13 give the per-replicate, per-channel numbers for each verification layer so external readers can audit them without rerunning the released scripts.

**Layer (i): GIS-grounded recomputation.** Table S10 compares the planner-reported deltas (from `mpc_summary.json`) against an independent recomputation that reads only the optimised DLTB shapefile, joins per-parcel slope from the prepared DEM raster, builds Queen contiguity via `libpysal`, and applies the same metric definitions as the main-text Methods. The recomputation script (`validate_optimized_shp.py`) shares no source with the optimisation pipeline.

**Layer (ii): ensemble member-subset ablation.** Table S11 reports the three replicates that constitute the variance estimator used in the main-text Methods caveat on five-seed determinism. Each replicate runs the full MPC pipeline ( $H=5$ ,  $K=50$ , greedy continuation, scoring=reward) with a different 2-of-3 ensemble member subset. The full ensemble’s result lies inside the mean  $\pm$  sd of the subset distribution on every reported metric.



Table S10: Independent GIS recomputation versus planner-reported deltas, Bishan run\_1, 53,086 swappable parcels. The recomputation invokes no learned model. Differences are at floating-point reduction noise level.

| Quantity                 | Recomputed (independent) | Planner-reported | $ \Delta $            |
|--------------------------|--------------------------|------------------|-----------------------|
| $\Delta$ slope (%)       | -2.0000                  | -2.0006          | $6 \times 10^{-4}$ pp |
| $\Delta$ contiguity      | +0.012739                | +0.012748        | $8 \times 10^{-6}$    |
| $\Delta$ baimu count     | +8                       | +8               | 0 (exact)             |
| $\Delta$ baimu area (ha) | -473.2950                | -473.2950        | $< 10^{-4}$ ha        |
| farm→forest swaps        | 428                      | 428              | 0 (exact)             |
| forest→farm swaps        | 428                      | 428              | 0 (exact)             |

Table S11: Per-replicate results for the member-subset ablation. The three replicates exhaust  $C(3,2)=3$  distinct subsets. Full-ensemble values reproduced from main-text Table 4 / `mpc_summary.json` for comparison; they fall inside the subset mean±sd on every metric.

| Replicate (members)   | $\Delta$ slope (%)   | $\Delta$ cont          | $\Delta$ baimu count | $\Delta$ baimu (ha) |
|-----------------------|----------------------|------------------------|----------------------|---------------------|
| {0, 1}                | -2.0222              | +0.01361               | +5                   | -475.02             |
| {0, 2}                | -1.9706              | +0.01437               | +5                   | -443.12             |
| {1, 2}                | -2.0024              | +0.01128               | +6                   | -505.39             |
| Mean ± sd (n=3)       | $-1.9984 \pm 0.0260$ | $+0.01309 \pm 0.00161$ | $+5.33 \pm 0.58$     | $-474.51 \pm 31.14$ |
| Full ensemble {0,1,2} | -2.0006              | +0.01275               | +8                   | -473.30             |

**Layer (iii): direct ensemble 1-step accuracy.** Table S12 reports the per-channel prediction error of the deployed 3-member ensemble against the true environment, accumulated over the 100 in-distribution steps that reproduce episode 0 of `mpc_summary.json`. The reward head is strongly under-fit but Spearman rank correlation between predicted and true reward is positive, and slope-channel error is sub-variance.

**Layer (iv): true-environment MPC counter-factual.** Table S13 compares the ensemble-MPC headline trajectory against an ablation that replaces every ensemble call with the actual env.step, under matched H/K/ $\gamma$  and a single seed. Implementation concessions (random rather than greedy continuation; stage-1 subsample of 200 of  $\sim 2,000$  valid actions) are forced by the cost of incremental simulation. Despite using a perfect dynamics model, the resulting trajectory is strictly worse on all three objectives.

Per-step trajectory matching between layer (iii) and the deployed pipeline is byte-identical at every monitored step (10, 20, ..., 100) of episode 0; this is the most stringent reproducibility check the verification stack supports and is therefore the canonical evidence that the deployed ensemble’s planning is deterministic given a fixed prepared dataset and config.

## S11 Policy translation: slope bands, township fairness, area trade-offs

This section reports the policy-relevant breakdown of the Bishan headline result that main-text §4 references. All numbers come from a single GIS-only post-hoc analysis script (`scripts/policy_translation_bishan.py`) that reads the optimised DLTB shapefile written by

Table S12: Ensemble 1-step prediction error over the same 100-step trajectory MPC visits (Bishan, seed 0). True-reward statistics are computed from the actual env.step return. Spearman rank correlation between predicted and true reward is +0.216. Pearson correlation between ensemble disagreement  $r_{\text{std}}$  and absolute prediction error is +0.077, i.e. disagreement does not flag where members are wrong.

| Quantity                                        | Value                         |
|-------------------------------------------------|-------------------------------|
| Reward MAE                                      | 0.818                         |
| Reward RMSE                                     | 1.540                         |
| Reward median  err                              | 0.170                         |
| Reward p90  err                                 | 3.148                         |
| Reward p99  err                                 | 5.075                         |
| $r_{\text{true}}$ range / std                   | $[-3.04, +6.36]$ / 1.397      |
| $r_{\text{pred}}$ range / std                   | $[+0.43, +0.67]$ / 0.069      |
| Naive constant-mean baseline MAE                | 0.828                         |
| Spearman ( $r_{\text{pred}}, r_{\text{true}}$ ) | +0.216                        |
| Ensemble $r_{\text{std}}$ mean / median / max   | 0.027 / 0.030 / 0.075         |
| Pearson ( $r_{\text{std}},  \text{err} $ )      | +0.077                        |
| $gf[4]$ slope-improvement MAE                   | 0.00238 (channel std 0.00521) |
| $gf[6]$ baimu-count-norm MAE                    | 0.00270 (channel std 0.00817) |

Table S13: True-environment MPC vs. deployed ensemble MPC, Bishan, 100 outer steps, seed 0. “True-env” replaces every ensemble call with actual env.step; otherwise H=5, K=50,  $\gamma=0.99$  unchanged. Implementation concessions stated in caption: random (rather than greedy) continuation, stage-1 subsample of 200 valid actions per step.

| Quantity                    | Ensemble MPC (run_1)        | True-env MPC             | $\Delta$ (true – ensemble) |
|-----------------------------|-----------------------------|--------------------------|----------------------------|
| $\Delta$ slope (%)          | −2.0006                     | −1.8187                  | +0.182 pp (worse)          |
| $\Delta$ contiguity         | +0.01275                    | +0.00865                 | −0.0041 (worse)            |
| $\Delta$ baimu count        | +8                          | +2                       | −6 (worse)                 |
| $\Delta$ baimu area (ha)    | −473.30                     | −565.10                  | −91.8 ha more loss         |
| Wall time (excl. env build) | 1,448 s                     | 358 s                    | true-env 4× faster         |
| Stage-1 coverage            | all valid actions (batched) | 200-action random subset | —                          |
| Stage-2 continuation        | greedy                      | random                   | —                          |

Tool 4 and shares no source with the optimisation pipeline; the per-band aggregates use parcel area in EPSG:32648 (UTM 48N) and the township assignment uses the first nine digits of QSDWM. The 13 Bishan core townships are the 500227xxx entries of the Tool 1 `townships.json` registry; the 8 500106xxx/500107xxx/500109xxx/500116xxx entries are Bishan-administered flying-territory parcels in neighbouring districts and are reported in the registry but excluded from the fairness summary.

**Slope-band shifts (Table S14).** The independent GIS recomputation of the on-disk optimised cadastre shifts 687 ha out of the  $\geq 15^\circ$  steep tail (the GB/T 21010-2017 marginal-cultivation band;  $-615$  ha from  $15\text{--}25^\circ$  and  $-72$  ha from  $> 25^\circ$ ), at the expense of a  $+182$  ha gain in the  $\leq 15^\circ$  mechanisable bands. The planner therefore executes *de facto* grain-for-green on the slope tail without explicit policy supervision, using the  $w_S$  slope reward as a proxy.

Table S14: Bishan farmland-area distribution across slope bands, before and after optimisation. Bands aligned with GB/T 30600-2022 (high-standard farmland mechanisability bands  $< 6^\circ$  and  $6\text{--}15^\circ$ ) and GB/T 21010-2017 (marginal-cultivation  $15\text{--}25^\circ$  and banned-cultivation  $> 25^\circ$ ). All areas in hectares, computed from the optimised DLTB shapefile under EPSG:32648 projection. Total farmland area in row 5 differs from row 1+row 2+row 3+row 4 of the “orig” / “opt” columns by  $\leq 0.1$  ha (rounding).

| Slope band                             | Original (ha) | Optimised (ha) | $\Delta$ (ha)            | Share orig $\rightarrow$ opt |
|----------------------------------------|---------------|----------------|--------------------------|------------------------------|
| $< 6^\circ$ (mechanisable)             | 15,892        | 15,940         | +48                      | 32.9% $\rightarrow$ 33.4%    |
| $6\text{--}15^\circ$ (terraceable)     | 28,187        | 28,320         | +134                     | 58.4% $\rightarrow$ 59.3%    |
| $15\text{--}25^\circ$ (marginal)       | 3,952         | 3,336          | $-615$                   | 8.2% $\rightarrow$ 7.0%      |
| $> 25^\circ$ (banned, GB/T 21010-2017) | 252           | 180            | $-72$                    | 0.5% $\rightarrow$ 0.4%      |
| <b>Net (cultivated land)</b>           | <b>48,283</b> | <b>47,776</b>  | <b><math>-505</math></b> | —                            |

**Township fairness (Table S15).** Per-township slope improvement on the 13 Bishan core townships ranges from  $-0.57\%$  (500227002, mean initial slope  $6.53^\circ$ ) to  $-4.08\%$  (500227100,  $9.85^\circ$ ); the cross-township mean is  $-2.07\%$  with sample standard deviation  $1.09\%$  and Gini coefficient  $0.28$  (computed on absolute slope-improvement magnitudes). No township is left at near-zero improvement, and the spread is consistent with the spread of initial slope means rather than with concentration of swaps in a small number of townships. The largest single-township farmland reduction is  $-90.2$  ha in 500227100, consistent with this township’s higher initial slope and per-parcel size.

**Area trade-off and the cultivated-land red line.** Two trade-offs are honestly reported (main-text Section 4): (i) total qualifying *baimu fang* area decreases by  $-473$  ha, equivalent to  $1.01\%$  of the initial  $46,844$  ha qualifying stock, a deliberate consequence of consolidating many small qualifying patches into fewer larger ones (the qualifying count rises by  $+8$  even as total qualifying area falls); (ii) the paired farm $\leftrightarrow$ forest swap conserves *counts* ( $426$  in,  $426$  out) but not *areas*, leading to a net cultivated-area decrease of  $-505$  ha ( $1.05\%$  of the  $48,283$  ha initial farmland stock). The two numbers are distinct: the qualifying-area loss is computed inside the  $\geq 6.67$  ha contiguous-farmland definition, the cultivated-land loss is computed across all farmland parcels. The cultivated-land red line is therefore not satisfied by the swap rule alone and must be enforced as an explicit constraint at deployment time. The practical implication is recorded in main-text Section 4 (Methods action paragraph) and main-text Discussion.

Table S15: Per-township breakdown across the 13 Bishan core townships. Slope means are area-weighted on farmland parcels; slope-change pct is computed against each township’s own initial slope. Farm→forest and forest→farm are counts of `CHG_FLAG = 1` and `CHG_FLAG = 2` parcels respectively. Source: `scripts/policy_translation_bishan.py`. The 8 flying-territory townships in the Tool 1 registry contribute zero swaps (`CHG_FLAG = 0` everywhere) and are omitted; their parcels carry through the optimisation unchanged.

| Town                  | Parcels | Init slope (°) | Opt slope (°) | $\Delta$ slope (%)        | Farm→For | For→Farm |
|-----------------------|---------|----------------|---------------|---------------------------|----------|----------|
| 500227001             | 3,906   | 9.540          | 9.383         | −1.65                     | 40       | 40       |
| 500227002             | 5,415   | 6.528          | 6.491         | −0.57                     | 13       | 13       |
| 500227100             | 2,317   | 9.853          | 9.450         | −4.08                     | 39       | 39       |
| 500227101             | 2,899   | 8.590          | 8.342         | −2.89                     | 33       | 33       |
| 500227102             | 5,875   | 9.646          | 9.507         | −1.44                     | 61       | 61       |
| 500227103             | 3,617   | 11.325         | 11.118        | −1.84                     | 49       | 49       |
| 500227104             | 2,538   | 11.310         | 11.169        | −1.24                     | 20       | 20       |
| 500227105             | 6,787   | 8.408          | 8.294         | −1.35                     | 27       | 27       |
| 500227106             | 5,739   | 7.220          | 7.157         | −0.88                     | 13       | 13       |
| 500227107             | 5,927   | 6.759          | 6.613         | −2.15                     | 36       | 36       |
| 500227108             | 2,669   | 7.905          | 7.591         | −3.98                     | 41       | 41       |
| 500227109             | 1,601   | 8.709          | 8.492         | −2.50                     | 18       | 18       |
| 500227200             | 3,225   | 6.364          | 6.214         | −2.37                     | 36       | 36       |
| <b>Range / spread</b> | —       | [6.36, 11.33]  | —             | [−4.08, −0.57], Gini 0.28 | —        | —        |

## S12 Hyperparameter sensitivity and longer-budget OR baselines

This section reports two robustness checks against reviewer concerns about hyperparameter dependence and the matched-wall-budget OR comparison: (i) an  $(H, K)$  sensitivity sweep on the package-side prepare and (ii) a 10×-extended wall-budget comparison against simulated annealing and random-restart on Bishan.

**$(H, K)$  sweep on the package-side prepare (Table S16).** We swept the planning horizon  $H \in \{3, 5, 10\}$  and the top- $K$  candidate width  $K \in \{25, 50, 100\}$  on the shipped contrastive ensemble, holding  $\lambda_{\text{rank}} = 5.0$ , margin = 0.1,  $\gamma = 0.99$ , continuation = greedy, scoring = reward, and one canonical episode per cell. All nine cells produced bit-identical commit sequences yielding the same on-disk slope improvement of −1.7531%, contiguity change of +0.0125, qualifying-count gain of +5, and qualifying-area loss of −483.94 ha. Wall time scales from 122.7 s ( $H=3, K=25$ ) to 1,139 s ( $H=10, K=100$ ), a 9.3× range. The invariance is consistent with the deterministic-planning property disclosed in main-text Methods (*Independent GIS-grounded verification*, transparency caveat 1): under fixed ensemble member ordering, top- $K$  rollout scoring, and greedy continuation, the contrastive ensemble’s batched 1-step  $r$ -prediction supplies a stable ranking signal whose top-1 commit is unchanged by deeper rollouts or wider candidate pools; the  $H$  and  $K$  knobs therefore trade only wall-clock against safety margin, not against quality, on a single ensemble. In practice we recommend ( $H=5, K=50$ ) as the default operating point and ( $H=3, K=25$ ) for an interactive scenario-exploration mode at  $\sim 2\times$  speed-up.

A note on the offset between −1.7531% here and the −2.001% on-disk verification of main-text Section 4: the sweep uses the `ensemble_seed0` member of the open per-seed checkpoint pack (released alongside the toolchain), while the headline ensemble in `mpc_summary.json` is the merged shipped ensemble at `prepared_legacy_horn/tool3/`. The two ensembles share the same

Table S16:  $(H, K)$  sensitivity on Bishan, package-side prepare, shipped contrastive ensemble (1 canonical episode per cell). All nine cells produce identical commit sequences; wall-clock cost is the only meaningful dimension on which they differ. Source: `scripts/sensitivity_sweep_bishan.py`, output `paper/submission_commsee/sensitivity_sweep_bishan.json`.

| $H$ | $K$ | Slope $\Delta$ (%) | $\Delta$ Cont. | $\Delta$ Baimu # | $\Delta$ Baimu (ha) | Reward | Wall-s  |
|-----|-----|--------------------|----------------|------------------|---------------------|--------|---------|
| 3   | 25  | -1.7531            | +0.0125        | +5               | -483.94             | 71.78  | 122.7   |
| 3   | 50  | -1.7531            | +0.0125        | +5               | -483.94             | 71.78  | 180.0   |
| 3   | 100 | -1.7531            | +0.0125        | +5               | -483.94             | 71.78  | 305.7   |
| 5   | 25  | -1.7531            | +0.0125        | +5               | -483.94             | 71.78  | 180.2   |
| 5   | 50  | -1.7531            | +0.0125        | +5               | -483.94             | 71.78  | 298.9   |
| 5   | 100 | -1.7531            | +0.0125        | +5               | -483.94             | 71.78  | 535.4   |
| 10  | 25  | -1.7531            | +0.0125        | +5               | -483.94             | 71.78  | 330.6   |
| 10  | 50  | -1.7531            | +0.0125        | +5               | -483.94             | 71.78  | 599.8   |
| 10  | 100 | -1.7531            | +0.0125        | +5               | -483.94             | 71.78  | 1,139.4 |

training pipeline ( $\lambda_{\text{rank}} = 5.0$ , margin = 0.1, 3 members, 30 epochs, identical hyperparameters; `train_summary.json seed_base = 0`) but were trained against slightly different transition data sources before and after the prepare-step rebuild; the gap (within the 0.7 pp band of the cross-pipeline reconciliation in §S3) is consistent with the per-ensemble variance reported in the open-source 5-ensemble reproduction ( $\sigma_{\text{slope}} \approx 0.024$ , mean -1.738%; §S3, “Cross-pipeline reproduction by the open-source toolchain”). The  $(H, K)$ -invariance result therefore characterises a single deterministic ensemble; we do not extrapolate it to ensembles trained from cold.

**Extended wall-budget OR baselines (Table S17).** To address the concern that a longer-budget classical baseline might catch up to learned-MPC, we re-ran simulated annealing and random-restart at a 10 $\times$ -longer 1,800s wall budget on Bishan (the same Bishan CountyLevelEnv, same env step cost of 14.3ms/step probed at run start). Both baselines run until the wall-clock budget expires; SA logs 1,187 iterations and random-restart logs 1,105 episodes. The reward gap to the contrastive-MPC pipeline persists by an order of magnitude, validating the simulator-cost crossover analysis of main-text Section 6.

Table S17: Bishan reward versus wall-clock budget across simulated annealing, random-restart, and contrastive MPC. Both OR baselines plateau at an order of magnitude below the contrastive-MPC headline regardless of compute; the additional 1,620 s of SA wall time buys only +0.50 reward units. Source: `runs/bishan/farmland_baselines_1800s.json` and `tests/farmland_baselines.py`.

| Method                  | Wall budget (s) | Reward            | Iterations / episodes | Notes                |
|-------------------------|-----------------|-------------------|-----------------------|----------------------|
| Random-action 5-ep mean | —               | $11.55 \pm 10.31$ | 5 episodes            | no method            |
| Simulated annealing     | 180             | 9.04              | 100 iter              | main-text Section 4  |
| Simulated annealing     | 1,800           | 9.54              | 1,187 iter            | this section (+5.5%) |
| Random-restart          | 180             | 30.85             | 98 ep                 | main-text Section 4  |
| Random-restart          | 1,800           | 37.70             | 1,105 ep              | this section (+22%)  |
| Contrastive MPC         | $\sim 180$      | <b>86.74</b>      | 1 ep, $H=5$ , $K=50$  | main-text Section 4  |